

Mutual Information Analysis of MLP Output and Residual Stream

Findings Summary for Team Discussion

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1 What We Computed

We measured the Gaussian mutual information between each layer’s representation and the final layer’s representation, using two different representations:

1. **MLP output $\mathbf{m}_\ell$** : the output of the MLP sub-module at layer ℓ , i.e., the additive contribution of that layer’s feedforward network.
2. **Residual stream $\mathbf{h}_\ell$** : the full hidden state after layer ℓ , which accumulates all prior contributions via skip connections: $\mathbf{h}_\ell = \mathbf{h}_{\ell-1} + \text{Attn}_\ell(\mathbf{h}_{\ell-1}) + \text{MLP}_\ell(\cdot)$.

For each, we compute:

$$I(\mathbf{z}_\ell; \mathbf{z}_L) \quad \text{where } \mathbf{z} \in \{\mathbf{m}, \mathbf{h}\}$$

under the assumption that $(\mathbf{z}_\ell, \mathbf{z}_L)$ are jointly Gaussian. This yields:

- **CCA method**: $I = -\frac{1}{2} \sum_i \log(1 - \rho_i^2)$, where ρ_i are canonical correlations from the whitened cross-covariance.
- **Log-det method**: $I = \frac{1}{2} (\log \det \Sigma_\ell + \log \det \Sigma_L - \log \det \Sigma_{\text{joint}})$.

Both formulas compute the same quantity (Gaussian MI) but differ in numerical stability.

1.1 Connection to Information Bottleneck

Following Tishby & Zaslavsky (2015), each layer can be characterized by two quantities in the *information plane*:

- $I(T_\ell; X)$ — how much the representation remembers about the input (proxied by **effective rank**).
- $I(T_\ell; Y)$ — how much the representation captures about the output (proxied by **Gaussian MI** with the final layer).

Our existing effective rank analysis provides the first axis; the MI analysis provides the second.

Model	Parameters	Layers	Hidden Size
Qwen3-0.6B	0.6B	28	1024
Qwen2.5-1.5B	1.5B	28	1536
SmolLM2-1.7B	1.7B	24	2048
Llama-3.2-1B	1.0B	16	2048

Table 1: Models used in the mutual information analysis. All results computed on 65,536 token activations from WikiText-2 training data.

2 Models Analyzed

3 Key Finding 1: MLP Output MI Shows Two-Phase Pattern with Drops

The MLP output MI (purple curves in Figure ??) shows a consistent two-phase pattern across all four models:

1. **Flat plateau** (early/middle layers): MI fluctuates in a relatively narrow band. These MLP sub-modules produce representations that are only moderately correlated with the final layer’s output.
2. **Exponential rise** (late layers, roughly final 30–40% of depth): MI increases steeply as each successive MLP’s output becomes increasingly aligned with the final representation.

3.1 The Drops: Near-Rank-1 MLP Layers

Every model exhibits sharp MI drops at specific layers. These drops correspond exactly to layers where the MLP output has **near-rank-1 effective rank** (effective rank ≈ 1 out of the full hidden dimension):

Model	Drop Layer(s)	Effective Rank	CCA Dims Used	MI (nats)
Qwen3-0.6B	2	1.004	1	3.75
2*Qwen2.5-1.5B	1	1.033	28	11.9
	2	1.047	31	11.6
	26	1.257	569	113.0
2*SmolLM2-1.7B	1	1.095	70	19.5
	7	1.119	5	1.43
Llama-3.2-1B	1	1.041	1442	∞

Table 2: Layers with near-rank-1 MLP output across all four models. “CCA Dims Used” indicates how many dimensions survived the truncated whitening threshold. Every drop in the MI curve corresponds to effective rank ≈ 1 .

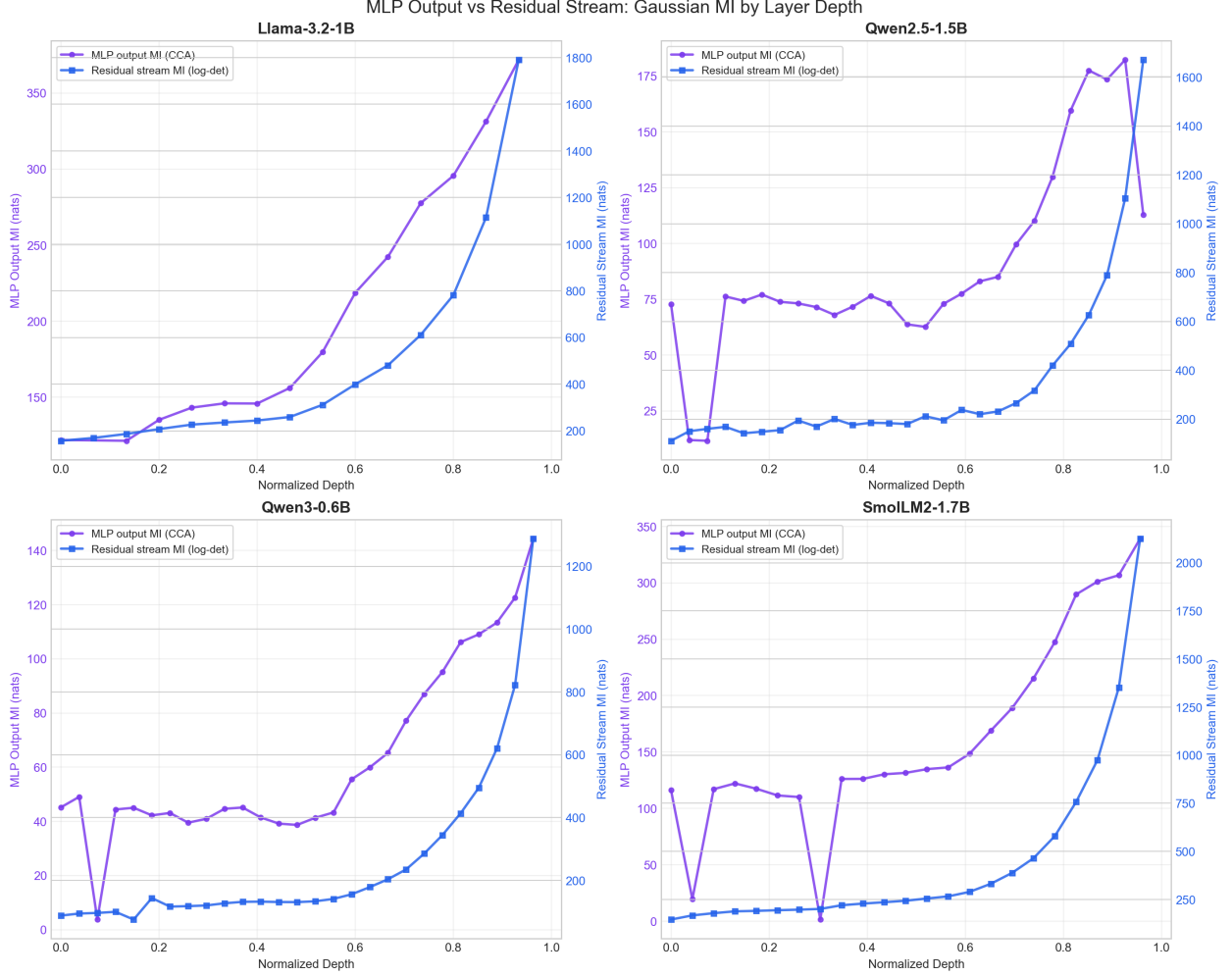


Figure 1: MLP output MI (purple, CCA method, left y-axis) vs. residual stream MI (blue, log-det method, right y-axis) across normalized layer depth for all four models. Note the dual y-axes: the absolute MI scales differ substantially between the two representations.

3.1.1 What does a rank-1 MLP output mean?

When a layer’s MLP output has effective rank ≈ 1 , it means the MLP’s contribution at that layer is essentially one-dimensional — all tokens’ MLP outputs are (approximately) scalar multiples of a single direction in the hidden space. This can happen because:

- The MLP learned a very specialized, low-rank transformation at that layer.
- The layer contributes almost nothing incrementally to the residual stream (its “contribution” is nearly constant or collinear).

A one-dimensional representation can carry at most a few nats of information about any target, regardless of how that information is measured. So the MI drop is **real in the sense that these MLPs genuinely contribute minimal information**, though the exact MI value is sensitive to the estimation method (CCA vs. log-det disagree substantially at these layers).

3.1.2 Why do CCA and log-det disagree at drop layers?

The CCA method requires *whitening* the covariance matrix ($\Sigma^{-1/2}$). For a rank-1 matrix, ~ 1000 near-zero eigenvalues get inverted, amplifying noise in the null space. Our truncated CCA mitigates this by only whitening in the subspace with significant variance, but the threshold choice affects the result.

The log-det method avoids whitening but adds regularization $\varepsilon \mathbf{I}$ to the covariance, which artificially inflates the determinant for near-singular matrices.

Neither method is fully reliable for near-rank-1 layers. We report CCA as the primary method and note these layers as measurement limitations.

3.1.3 Implication for compression

These rank-1 layers are the **most compressible layers** by every measure:

- Lowest effective rank (minimal representation complexity, low $I(T; X)$).
- Lowest MI with final output (minimal informativeness, low $I(T; Y)$).
- Removing or heavily compressing these MLPs should have minimal impact on model output.

This is consistent with the Information Bottleneck prediction: layers that are maximally compressed while retaining minimal task-relevant information sit at the origin of the information plane.

4 Key Finding 2: Residual Stream MI Is Smooth and Monotonic

The residual stream MI (blue curves) is smooth and monotonically increasing for all four models. This is theoretically expected: the residual stream forms a Markov chain $X \rightarrow \mathbf{h}_0 \rightarrow \mathbf{h}_1 \rightarrow \dots \rightarrow \mathbf{h}_L$, and by the Data Processing Inequality, $I(\mathbf{h}_\ell; \mathbf{h}_L)$ must increase with ℓ .

The contrast between the two curves is informative:

- **Where MLP output MI is low but residual stream MI increases:** the layer’s MLP contributes little new information — the increase comes from what was already in the residual stream via skip connections. These are *redundant* MLP layers.
- **Where MLP output MI rises sharply:** the layer’s MLP is making a significant, new contribution to the final representation. These are *critical* MLP layers.
- **At rank-1 drop layers:** the residual stream continues its smooth increase, confirming that *no information is lost* in the overall network — the skip connection carries everything past the degenerate MLP.

5 Key Finding 3: Universal Pattern Across Architectures

Despite different architectures, parameter counts, and training procedures, all four models share:

1. **MLP output MI:** flat early $\rightarrow$ exponential late growth, with 1–3 rank-1 anomaly layers.
2. **Residual stream MI:** smooth monotonic increase, accelerating in the final $\sim 30\%$ of depth.
3. **Rank-1 MLP layers exist in every model**, typically in the first few layers. Qwen2.5-1.5B additionally has one near the final layer (layer 26).

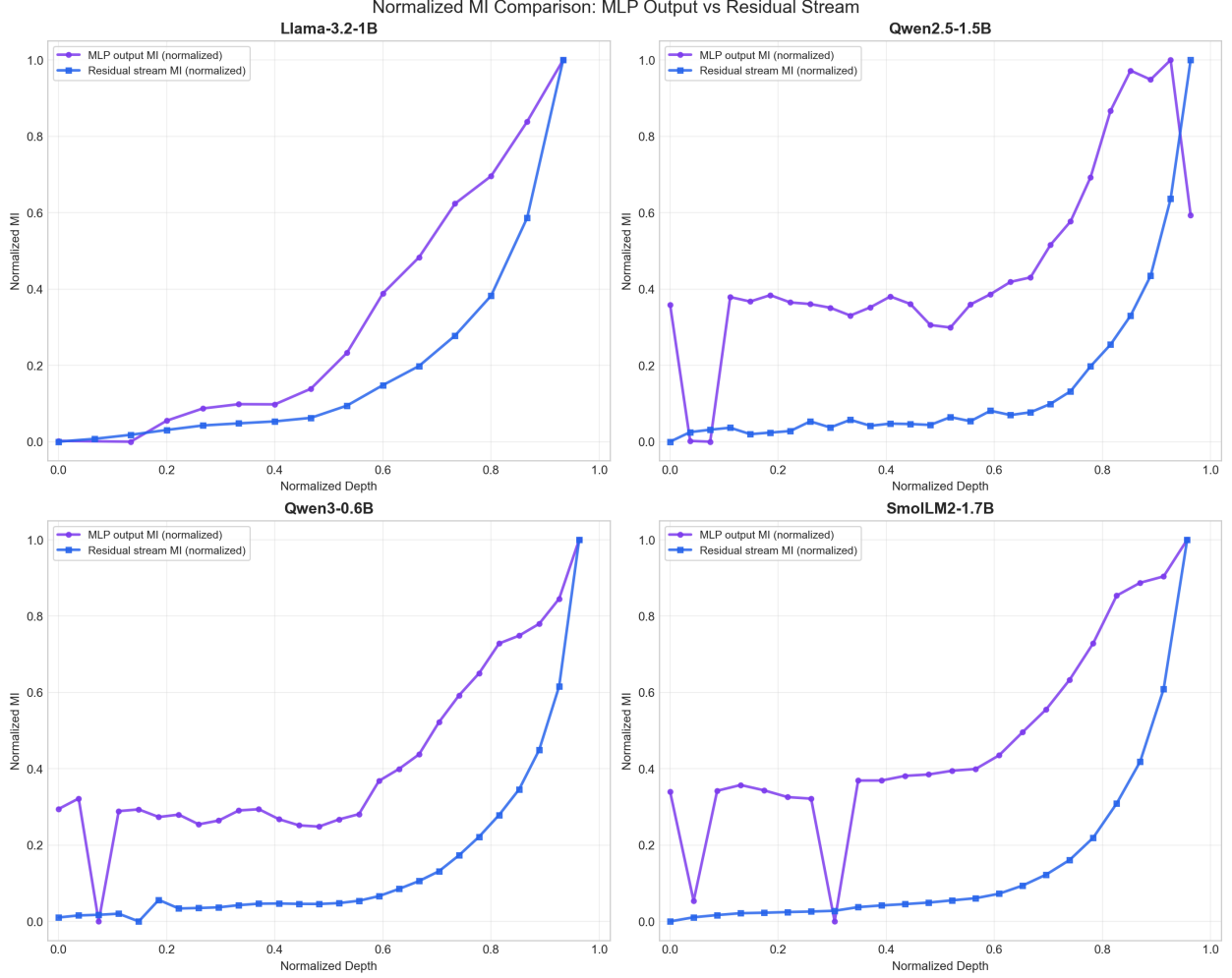


Figure 2: Normalized MI comparison (both min-max scaled to $[0, 1]$). Purple: MLP output (CCA). Blue: residual stream (log-det). The residual stream shows smooth, monotonically increasing MI across all models, while MLP output shows drops at rank-1 layers and a flatter early/middle profile.

4. **The exponential growth region** in MLP output MI aligns with the region where effective rank tends to decrease — consistent with the IB tradeoff (late layers compress while becoming more informative).

6 Methodology Notes

6.1 Why Gaussian MI?

MLP outputs pass through nonlinear activations (SiLU/GELU) and are not truly Gaussian. However:

- Gaussian MI provides a **lower bound** on true MI (among distributions with the same covariance, Gaussian has minimum MI).
- Non-parametric MI estimators (KNN, binning) do not work in 1024+ dimensions with $\sim 65k$ samples.

- The **relative ordering** between layers is reliable even under the Gaussian approximation.

6.2 MLP output vs. residual stream: which to use for IB?

- **MLP output**: answers “how informative is this layer’s individual contribution?” Directly connects to per-layer compression decisions.
- **Residual stream**: answers “how informative is the cumulative representation at this point?” Matches the Tishby–Zaslavsky information plane framework.
- For the IB information plane, we recommend using **effective rank of MLP output** as $I(T; X)$ and **residual stream MI** as $I(T; Y)$, since the residual stream provides a stable, monotonic measure of cumulative informativeness.

6.3 Estimation methods

- **MLP output MI**: CCA with truncated whitening (primary). Eigenvalues below $10^{-5} \times \lambda_{\max}$ are excluded before whitening to avoid noise amplification in the null space.
- **Residual stream MI**: Log-det (primary). More stable than CCA on the residual stream because CCA also produces ∞ at layers with low residual effective rank.
- Both methods compute the same Gaussian MI; the choice is driven by numerical stability for each representation type.

7 Summary for Team

1. We now have two MI measurements per layer: **MLP output** (per-layer contribution) and **residual stream** (cumulative representation). Together with **effective rank**, this gives us the full IB picture.
2. The **drops in MLP output MI** are real — they correspond to layers whose MLP produces near-rank-1 output (essentially one-dimensional). These are the most compressible layers.
3. The **residual stream MI is smooth and monotonic**, confirming that skip connections preserve information past degenerate MLP layers.
4. The **exponential MI growth in late layers** combined with **declining effective rank** in the same region is empirical evidence for the Information Bottleneck tradeoff in trained LLMs.
5. All four models show the same qualitative pattern, suggesting this is a **universal structural property** of transformer-based language models.

References

- [1] N. Tishby and N. Zaslavsky, “Deep Learning and the Information Bottleneck Principle,” *IEEE Information Theory Workshop (ITW)*, 2015. arXiv:1503.02406.
- [2] R. Shwartz-Ziv and N. Tishby, “Opening the Black Box of Deep Neural Networks via Information,” arXiv:1703.00810, 2017.